BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A

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TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

WRITTEN ASSESSMENT

TIME: 2 HOURS Cat 1

INSTRUCTIONS TO THE CANDIDATE

1. *Answer all questions.*
2. *Write your answers clearly and legibly.*
3. *Show all working where calculations are required.*
4. *Where diagrams are required, use appropriate electrical/electronic symbols.*
5. *The marks allocated to each question are shown in brackets.*

1.

a) Define an electrical circuit. (2 marks)

b) State four components of a basic electrical circuit. (4 marks)

c) State the SI unit for each of the following electrical quantities: (5 marks)

Electrical quantity	SI unit
EMF	
Current	
Power	
Resistance	
Energy	

2.

a) What is meant by electric current? (2 marks)

b) A resistor has a resistance of 10 Ω and carries a current of 2 A. Calculate the voltage across the resistor. (3 marks)

c) Distinguish between AC and DC. (2 marks)

application of DC. (1 mark)

3.

a) Determine the resistance value of a resistor with the following colour bands:

Brown – Black – Red – Gold. (3 marks)

b) Determine the resistance value and tolerance of a resistor with the following colour bands:

Red – Violet – Orange – Gold. (3 marks)

c) State the function of the fourth colour band on a resistor. (2 marks)

4.

a) Define a semiconductor. (2 marks)

b) Name two semiconductor materials. (2 marks)

c) With an aid of a diagram differentiate between P-type and N-type semiconductor. (4 marks)

d) State the two biasing conditions of a PN-junction diode. (2 marks)

5.

a) With the aid of a labelled V–I characteristic curve, describe the behaviour of a PN-junction diode in forward bias and reverse bias. (6 marks)

b) State two applications of a PN-junction diode. (2 marks)

6.

a) What is a transistor? (2 marks)

b) Name the two types of bipolar junction transistors. (2 marks)

c) State the three terminals of a bipolar junction transistor. (3 marks)

d) Give three applications of transistors in electronic circuits. (3 marks)

7.

a) Define an integrated circuit (IC). (2 marks)

b) State three advantages of using integrated circuits in electronic equipment. (3 mark

c) State one characteristic of each of the following components: (4 marks)

i. Resistor

ii. Capacitor

iii. Diode

iv. Inductor

d) Give one application of an integrated circuit. (1 mark)

8.

a) Define computer memory. (2 marks)

b) State the two main classifications of computer memory. (2 marks)

c) Give two examples of primary memory. (2 marks)

d) Give two examples of secondary storage. (2 marks)

e) State two differences between primary memory and secondary storage. (2 marks)

9.

a) Arrange the following memory levels from fastest to slowest: (5 marks)

- Secondary storage
- Registers
- Main memory
- Tertiary storage
- Cache memory

b) State one characteristic of each of the following: (5 marks)

i. Registers

ii. Cache memory

iii. Main memory

iv. Secondary storage

v. Tertiary storage

10.

a) State the difference between RAM and ROM. (4 marks)

b) State two characteristics of RAM. (2 marks)

c) State two characteristics of ROM. (2 marks)

d) Classify computer memory according to the technology used by giving examples for each: (2 marks)

i. Semiconductor memory — _____

ii. Magnetic memory — _____

iii. Optical memory — _____

1a) Electrical circuit is a complete conducting path through which electric current flows from the source through electrical components and back to the source.

b) Electrical energy source.
Conductor
Load
Switch

c)

Electrical quantity	SI unit
EMF	volts (V)
Current	Amperes (A)
Power	Watt (W)
Resistance	Ohms (Ω)
Energy	Joules (J)

2. Electric current is the flow of electric charge through a conductor or circuit.

b) $V = IR$

$I = 2A$

$R = 10\Omega$

$10 \times 2 = 20$

$= 20$ volts

g) An alternating current changes direction periodically while a direct current flows in one direction.

d) Powering battery-operated devices.

3. Brown - 1
Black - 0
Red - 2
Gold - +/-5%

$10 \times 10^2 \pm 5\%$

$10 \times 10^2 = 1000$

$5\% \times 1000 = 50$

1000 ± 50 ohms.

b) Red - 2
Violet - 7
Orange - 3
Gold ± 5%

$27 \times 10^3 \pm 5\%$

27000

$\frac{5}{100} \times 27000 = 1350$

= 13500hms.

Find min value & max value

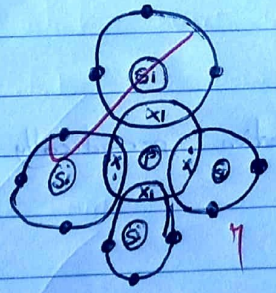
c) It indicates the tolerance of the resistor.

4a) Semi-conductor is a material whose electrical conductivity that lies between that of a conductor and an insulator.

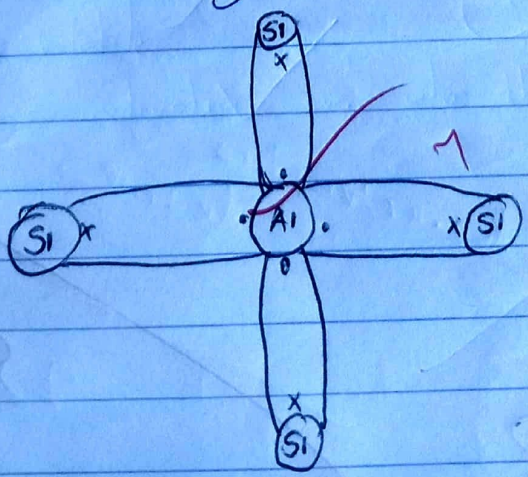
b) Silicon
Germanium.

d) ~~P Type~~

N-type



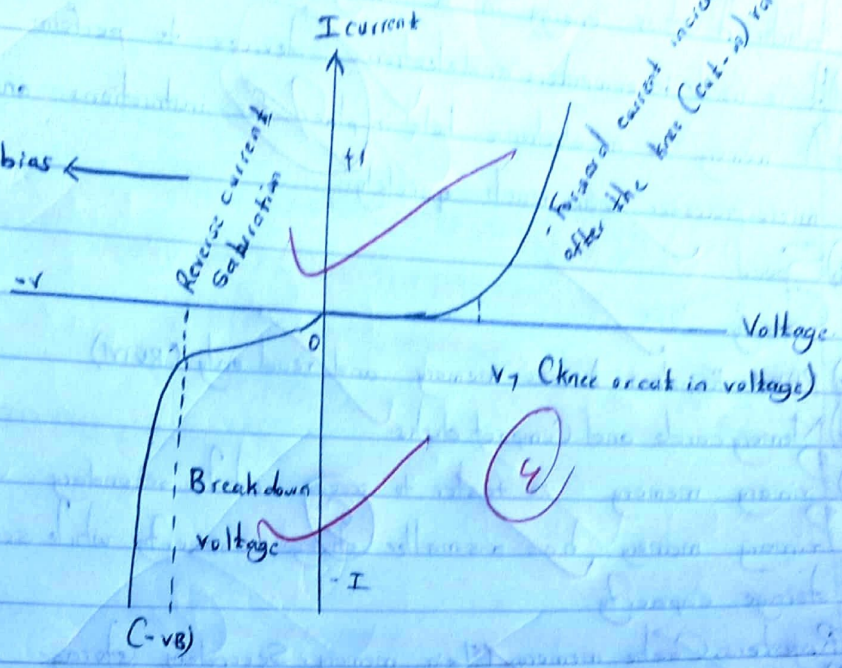
P-type



Are bonds formed.

Forward biasing.
Reverse biasing (2)

5a)



(2)

b) Rectification
Switching. (2)

6a) A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power. (2)

b) NPN transistor.
PNP transistor. (2)

c) Emitter (E)
Base (B)
Collector (C) (3)

d) Amplification of electrical signals.
Switching in electronic circuits.
Oscillation for generating electrical ~~components~~ signals. (3)

7a) An Integrated circuit is a small electronic device in which many ele components are manufactured and interconnected on a single semiconductor chip. (2)

b) They are small in size.
They have low power consumption. (3)
It is reliable.

c) Resistor - it limits the flow of electric current.
Capacitor - it stores electrical charge.

Characteristics not define

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iii) Diode - it allows electric current to flow mainly in one direction.
Inductor - it stores energy in a magnetic field when current flows thrpugh it.

7d) It is used in computers and electronic devices to perform processing and control functions.

8a) A memory is an electronic hold in place for instructions and data that your computer's micro-processor can reach quickly.

b) Speed
Capacity

c) RAM - Random Access memory and read only (ROM)

d) Memory cards and Compact disks

e) Primary memory is faster to access while secondary storage is slower to access.
Primary memory has a smaller storage capacity while secondary storage has a l
storage capacity.

9a) Registers, Cache memory, Main memory, Secondary storage and tertiary storage.

b) Registers - It has the fastest memory.
Cache memory - it stores frequently used data and instructions.
Main memory - it holds programs and data currently being used by the com
Secondary storage - it provides long-term storage of data.
Tertiary storage - it is used for large-scale backup.

10a) RAM is volatile while ROM is non-volatile.
RAM losses it's contents when power is switched off while ROM does not lose its contents.

b) Characteristics of RAM.
It is volatile.
It allows both reading and writing.
Characteristics of ROM
It is non volatile.
It is stores instructions even when the power is switched off.

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